ANDREA VILLARRUEL GARCÍA

Genomic Sciences Undergraduate Student

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ACADEMIC PROFILE

Genomic Sciences undergraduate with hands-on laboratory experience in bacterial genetics and quorum sensing, alongside computational training through undergraduate projects in transcriptomics and genome assembly. Interested in cellular differentiation, epigenetics, and evolutionary genomics, particularly at the interface of experimental and computational biology.

EDUCATION

B.Sc. in Genomic Sciences Aug 2024 – Jun 2028 (expected)

National Autonomous University of Mexico (UNAM) · Center for Genomic Sciences (CCG)

Overall academic average: 8.58/10

Selected coursework: Bioinformatics; Molecular Biology; Cell Biology; Epigenetics; R and Statistics; Genome Assembly and Annotation; Python; Differential Equations; Evolution; Data Science.

RESEARCH AND ACADEMIC EXPERIENCE

Undergraduate Research Student Jan 2025 – Present

Evolutionary Genomics Laboratory · CCG-UNAM · Advisor: Dr. Miguel Ángel Cevallos Gaos

- Contribute to research examining N-acyl homoserine lactone (AHL) signaling across *Rhizobium* strains.
- Perform T-streak bioassays using an *Agrobacterium tumefaciens* biosensor to detect AHL-associated activity through colorimetric readouts.
- Perform genomic DNA extraction, PCR and colony PCR, restriction digestion, DNA ligation, bacterial transformation, molecular cloning, and agarose gel electrophoresis.
- Maintain bacterial cultures; perform antibiotic selection and IPTG/X-Gal blue-white screening.

Stem Cell Differentiation Protocol Curation (Remote Academic Collaboration) Aug 2026 – Present

Leiden University Medical Center (LUMC) · Research group of Prof. Dr. Micha Drukker

- Review peer-reviewed stem cell literature and extract experimental details from differentiation protocols.
- Organize culture conditions, reagents, timelines, and readouts in structured JSON records for a research protocol platform; revise entries following methodological review and feedback.

SELECTED ACADEMIC PROJECTS

Mouse Telencephalon RNA-seq Reanalysis | Transcriptomics course project, 2026 | Repository

- Reanalyzed public bulk RNA-seq data (E11.5 vs. E14.5) using STAR, featureCounts, DESeq2, and enrichment analyses; identified 3,273 differentially expressed genes.

De Novo Assembly and Annotation of a Bacterial Genome | Collaborative course project, 2026 | Repository

- Collaboratively compared bacterial genome assemblies from Illumina and PacBio reads using SPAdes, Unicycler, Canu and Flye; evaluated assemblies with QUAST, polished with Pilon, and annotated with Prokka.

TECHNICAL SKILLS

Experimental: Bacterial culturing and transformation; AHL biosensor assays; antibiotic selection; PCR/colony PCR; DNA extraction; restriction digestion, ligation and cloning; agarose gel electrophoresis; preparation of loading buffers, IPTG and X-Gal solutions.

Computational: R, Bash, Python (scripting); RNA-seq (STAR, featureCounts, DESeq2); genome assembly (SPAdes, Canu, Flye, Pilon); QUAST, BLAST, Prokka; Git/GitHub.

LANGUAGES

Spanish: Native | **English:** B2 (TOEFL ITP)